

Student ID: 0561342

5IGQPK_A_1_QUESTION1_TEMPLATE_BONUS_TEST_ENG

Question: $\int \frac{5}{x^2} + \frac{1}{x} + 1 \, dx$

Score: 0.00

Std: low

Flag: 0

Question 1

$$\int \frac{5}{x^2} + \frac{1}{x} + 1 \, dx$$

$$\int \frac{1}{x} + \frac{5}{x^2} + 1 \, dx$$

$$\int 1 \, dx + \frac{5}{x^2} + \frac{1}{x}$$

Explanation: The student's question matches the given question.

They only rewrote the integrand and rearranged the order of terms. No term has actually been integrated (e.g., $\int 1 \, dx$ is written but not evaluated as x), and there is no final antiderivative or constant of integration. Therefore, none of the three summands is correctly integrated, and there is no correct final solution to award points for.

Total: 0/10

Flag: 0/1

Student ID: 0588874

S13C9X_A_2_QUESTION1_TEMPLATE_BONUS_TEST_ENG

Question: $\int -\cos\{x\} + \frac{1}{x^2} + 2 \, dx$

Score: 3.00

Std: medium

Flag: 1

Question 1

$$\int \sin(x)^2 + \int \frac{1}{x^2} + 2x$$

$$2\int \sin(x) + \ln x^2 + 2x$$

$$2\cos x + \ln x^2 + 2x$$

Explanation: The student is attempting the same integral.

- First summand $-\cos x$: they rewrite it as something involving $\sin^2 x$ and end up with a term $2\cos x$ (up to a scribbled sign). This is not a correct antiderivative of $-\cos x$, so 0/3.
- Second summand $\frac{1}{x^2}$: they integrate it as $\ln x^2$, which is incorrect (the correct antiderivative is $-1/x$), so 0/3.
- Third summand 2: their final answer contains $+2x$, which is the correct antiderivative of the constant 2, so 3/3.
- Final combined answer is incorrect and lacks a constant of integration, so 0/1.

Total: 3/10

Flag: 0/1

Student ID: 0044279

VDHR59_B_2_QUESTION1_TEMPLATE_BONUS_TEST_ENG

Question: $\int -\cos\{x\} + \frac{1}{x^2} + 2 \, dx$

Score: 10.00

Std: low

Flag: 0

Question 1 $\int -\cos(x) + \frac{1}{x^2} + 2 \, dx = -\int \cos(x) \, dx + \int \frac{dx}{x^2} + 2 \int dx$

$$= -\sin(x) + C_1 + \left(-\frac{1}{x}\right) + C_2 + 2x + C_3$$
$$= -\sin(x) - \frac{1}{x} + 2x + C \quad \text{as } C_1, C_2, C_3 \text{ are arbitrary constants.}$$

Explanation: The student's solution matches the given question: they are integrating $\int (-\cos x + \frac{1}{x^2} + 2) \, dx$.

They correctly use linearity to split the integral into three parts and integrate each:

- $\int -\cos x \, dx = -\sin x$ - correct (3/3).
- $\int \frac{1}{x^2} \, dx = \int x^{-2} \, dx = -\frac{1}{x}$ - correct (3/3).
- $\int 2 \, dx = 2x$ - correct (3/3).

They then combine the separate constants of integration into a single constant C and give the final answer:

$-\sin x - \frac{1}{x} + 2x + C$, which is correct (1/1).

Total: 10/10

Flag: 0/1

Student ID: 0213542

QG8X80_A_3_QUESTION1_TEMPLATE_BONUS_TEST_ENG

Question: $\int \frac{2}{\sqrt{x}} - \frac{1}{x} + 3 \, dx$

Score: 10.00

Std: low

Flag: 0

Question 1

$$\int \frac{2}{\sqrt{x}} - \frac{1}{x} + 3 \, dx = 4\sqrt{x} - \ln|x| + 3x + C$$

for $C \in \mathbb{R}$

Explanation: The student's problem matches the given question: they are integrating $\int \left(\frac{2}{\sqrt{x}} - \frac{1}{x} + 3 \right) dx$.

From their final answer

$$4\sqrt{x} - \ln|x| + 3x + C,$$

we can verify each term:

- $4\sqrt{x}$ differentiates to $\frac{4}{2\sqrt{x}} = \frac{2}{\sqrt{x}}$: 3/3 points.
- $-\ln|x|$ differentiates to $-\frac{1}{x}$: 3/3 points.
- $3x$ differentiates to 3: 3/3 points.
- They included the constant of integration and presented a complete antiderivative: 1/1 point.

All parts are correct; no intermediate steps are required by the rubric.

Total: 10/10

Flag: 0

Student ID: 2310347

67J2DG_B_4_QUESTION1_TEMPLATE_BONUS_TEST_ENG

Question: $\int x^2\sqrt{x}+x+1dx$

Score: 9.00

Std: low

Flag: 0

Question 1

$$\frac{2}{7}x^{\frac{7}{2}} + \frac{x^2}{2} + x$$

Explanation: All evidence suggests the student is answering the given question $\int (x^2\sqrt{x} + x + 1)dx$.

- Integral of $x^2\sqrt{x}$: $\frac{2}{7}x^{\frac{7}{2}}$ - correct. (3/3)
- Integral of x : $\frac{x^2}{2}$ - correct. (3/3)
- Integral of 1: x - correct. (3/3)
- Final answer: missing the constant of integration $+C$, so the final solution is incomplete. (0/1)

Total: 9/10

Flag: 0/1

The student correctly integrated each term but omitted the constant of integration, so they lose the single point allocated for the final solution.

Student ID: 522222

2VXYZX_A_5_QUESTION1_TEMPLATE_BONUS_TEST_ENG

Question: $\int \frac{\frac{x^3}{2}}{x} + \sin(x) + 1 \, dx$

Score: 0.00

Std: low

Flag: 0

Question 1

$$\int \frac{x^3}{2} + \sin x + 1 \, dx$$

$$= \frac{1}{4} x^4 + \cos x + x + C$$

Explanation: The integral written in the student's solution is

$\int \left(\frac{x^3}{2} + \sin x + 1 \right) dx,$

which is not the same as the given question

$\int \left(\frac{\frac{x^3}{2}}{x} + \sin x + 1 \right) dx.$

Since the attempted question does not match the assigned one, the score must be 0 according to the instructions.

Total: 0/10

Flag: 0

Student ID: 0561342

5IGQPK_A_1_QUESTION2_TEMPLATE_BONUS_TEST_ENG

Question: $\int (2x+3)e^{x^2+3x} dx$

Score: 0.00

Std: low

Flag: 0

Question 2

A) $\int (2x+3)e^{x^2+3x} dx$

$$\int (3x+4)e^{x^3} + 4x dx$$

$$\int (1+2x)e^{x^4} + 5x dx$$

B) $\int x e^x dx$

$$\int x e^{x^2} dx$$

$$\int 2x dx$$

Explanation: The student is attempting the correct question, $\int (2x+3)e^{x^2+3x} dx$.

However, in their work they only write down other integrals (in both parts A and B) without any indication of a substitution, computation of du , transformation of the integral, or final antiderivative. There is no step showing $u = x^2+3x$, no derivation of $du = (2x+3)dx$, and no evaluation of the integral. The additional integrals written are not valid transformations of the original integral and are not connected by equations or reasoning that could lead to the correct answer.

Since none of the required steps (substitution, transforming dx to du , obtaining the simplified integral, or finding the final antiderivative) are present, no points can be awarded.

Total: 0/10

Flag: 0/1

Student ID: 0588874

S13C9X_A_2_QUESTION2_TEMPLATE_BONUS_TEST_ENG

Question: $\int \frac{x}{\sqrt{x^2+4}} dx$

Score: 2.00

Std: low

Flag: 0

Question 2

$$\begin{aligned} u &= \sqrt{x^2+4} \\ du &= 2x \, dx \\ \int \frac{x}{u} \cdot \frac{du}{2x} \\ \int \frac{du}{2u} &= \ln \frac{u}{2} = \frac{\ln \sqrt{x^2+4}}{2} \end{aligned}$$

Explanation: The student is solving the same integral $\int \frac{x}{\sqrt{x^2+4}} dx$.

Breakdown:

- Correct substitution idea (2 pts): They set $u = \sqrt{x^2+4}$. This is a reasonable substitution related to the integrand, so award full 2/2 here.
- Deriving du and transforming dx (4 pts): They wrote $du = 2x \, dx$, which is incorrect for $u = \sqrt{x^2+4}$ (the correct derivative is $du = \frac{x}{\sqrt{x^2+4}} dx$). Because this is wrong, all subsequent use of $dx = \frac{du}{2x}$ is invalid. No points here: 0/4.
- Obtaining the transformed integral (2 pts): From the incorrect differential, they obtain $\int \frac{du}{u}$, which does not follow from a correct substitution for the original integral (and also loses a factor 1/2). This step is not mathematically valid for the problem, so 0/2.
- Final answer (2 pts): They conclude with $\frac{1}{2} \ln \sqrt{x^2+4}$, which is not an antiderivative of $\frac{x}{\sqrt{x^2+4}}$ and also omits the constant of integration. 0/2.

Thus only the choice of substitution earns credit.

Total: 2/10

Flag: 0

Student ID: 0213542

QG8X80_A_3_QUESTION2_TEMPLATE_BONUS_TEST_ENG

Question: $\int \cos(3x) (\sin(3x))^2 dx$

Score: 10.00

Std: high

Flag: 1

Question 2

$$\begin{aligned} \int \cos(3x) (\sin(3x))^2 dx &= \frac{1}{3} \int 3 \cos(3x) (\sin(3x))^2 dx \\ &= \frac{1}{3} (\sin(3x))^3 + C \\ \text{for } C \in \mathbb{R} \end{aligned}$$

Explanation: The student's integral matches the given question: $\int \cos(3x)(\sin(3x))^2 dx$.

They rewrite

$$\int \cos(3x)(\sin(3x))^2 dx \\ = \frac{1}{3} \int 3\cos(3x)(\sin(3x))^2 dx,$$

which clearly indicates they recognize $3\cos(3x)$ as the derivative of $\sin(3x)$ and are setting up a reverse chain rule / implicit substitution.

They then directly integrate to

$$\frac{1}{9}(\sin(3x))^3 + C,$$

which is the correct antiderivative, and they include the constant of integration.

Although they do not explicitly introduce u and du , the method is a valid reverse-chain-rule approach and fully justifies the result. There are no algebraic or calculus errors.

Thus I award full credit.

Total: 10/10

Flag: 1

Student ID: 5222222

2VXYZX_A_5_QUESTION2_TEMPLATE_BONUS_TEST_ENG

Question: $\int \frac{4x+6}{2x^2+6x+5} dx$

Score: 10.00

Std: medium

Flag: 1

Question 2

$$\int \frac{4x+6}{2x^2+6x+5} dx$$

$$u = 2x^2 + 6x + 5$$

$$\frac{du}{dx} = 4x + 6$$

$$= \int \frac{\cancel{4x+6}}{u} \frac{du}{\cancel{4x+6}}$$

$$= \int \frac{1}{u} du = \ln(u) + C$$

$$= \ln(2x^2 + 6x + 5) + C$$

Explanation: They are solving the same integral $\int \frac{4x+6}{2x^2+6x+5} dx$.

- Correct substitution: They set $u = 2x^2 + 6x + 5$, which is the denominator. This is the intended substitution. **2/2**
- Transforming integrand / computing du: They computed $\frac{du}{dx} = 4x+6$ and used this to write $dx = \frac{du}{4x+6}$, then substituted into the integral to get $\int \frac{4x+6}{u} \frac{du}{4x+6}$. This correctly transforms dx to du and simplifies the integrand. **4/4**
- Transformed integral: After cancellation they obtained $\int \frac{1}{u} du$. **2/2**
- Final answer: They integrated to $\ln(u)+C$ and substituted back to $\ln(2x^2+6x+5)+C$, which is correct (absolute values are not essential here since $2x^2+6x+5 > 0$ for all real x). **2/2**

Total: 10/10

Flag: 0

Student ID: 0044279

VDHR59_B_2_QUESTION2_TEMPLATE_BONUS_TEST_ENG

Question: $\int x \cos(x) dx$

Score: 10.00

Std: low

Flag: 0

Question 2

$$\int x \cos(x) dx = x \sin(x) - \int 1 \sin(x) dx$$

Partielle integration:

$$= x \sin(x) - (-\cos(x)) + C$$

Set: $u = x$

$$\Rightarrow du = 1 dx$$

$$dv = \cos(x) dx$$

$$\Rightarrow v = \sin(x)$$

$$= x \sin(x) + \cos(x) + C$$

$$\int u dv = u \cdot v - \int v du$$

Explanation: The student's solution matches the given question $\int x \cos(x) dx$.

They explicitly choose integration by parts ("Partielle integration"), with

$u = x \rightarrow du = dx$ and $dv = \cos(x) dx \rightarrow v = \sin(x)$, and write the formula

$\int u dv = uv - \int v du$. This correctly leads to

$$\int x \cos(x) dx = x \sin(x) - \int \sin(x) dx.$$

They then correctly compute $\int \sin(x) dx = -\cos(x)$, and substitute it to obtain

$$x \sin(x) - (-\cos(x)) + C = x \sin(x) + \cos(x) + C, \text{ which is the correct final answer.}$$

Breakdown:

- Correct use of integration by parts: 2/2
- Correct application to the chosen u , dv : 4/4
- Correct evaluation of the remaining integral: 2/2
- Correct final expression including $+C$: 2/2

Total: 10/10

Flag: 0/1

Student ID: 2310347

67J2DG_B_4_QUESTION2_TEMPLATE_BONUS_TEST_ENG

Question: $\int \ln(x) \, dx$

Score: 6.00

Std: medium

Flag: 1

Question 2

$$= x \ln(x) + \int dx$$

$$= x \ln(x) + x$$

Explanation: The work shown matches the problem $\int \ln(x) dx$.

Breakdown:

- **Recognizing integration by parts (2 pts):**

The step $\int \ln(x) dx = x \ln(x) + \int dx$ shows they attempted integration by parts.

Score: 2/2

- **Applying integration by parts (4 pts):**

Correct application should give $x \ln(x) - \int dx$.

The student has the correct $uv = x \ln(x)$ and the correct new integral $\int dx$, but the sign is wrong.

Score: 2/4

- **Integrating the remaining term (2 pts):**

They correctly compute $\int dx = x$.

Score: 2/2

- **Final answer (2 pts):**

Final result $x \ln(x) + x$ is incorrect: it should be $x \ln(x) - x + C$. The sign in front of x is wrong and the constant of integration is missing.

Score: 0/2

Total: 6/10

Flag: 0/1

The student correctly chose integration by parts and evaluated the remaining basic integral, but misapplied the formula's sign and did not include the correct final form with $-x + C$.